

DONGWHEE KIM

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SUMMARY

I am a Ph.D. student at The University of Texas at Austin, specializing in computer architecture with a focus on improving the reliability of quantum computers and memory systems. My academic background includes research presented at leading conferences such as HPCA and SC. Prior to my Ph.D. studies, I worked as an HBM I/O Circuit Design Engineer at Samsung Electronics and earned both my B.S. and M.S. degrees from Sungkyunkwan University.

- **Research Interests:** Computer Architecture, Quantum Computing, Memory Systems, Reliability

EDUCATION

The University of Texas at Austin (UT Austin) <i>Ph.D., Electrical and Computer Engineering</i> Advisor: Prof. Poulami Das	Texas, United States <i>Aug. 2025 – Present</i>
Sungkyunkwan University (SKKU) <i>M.S., Semiconductor and Display Engineering</i> - Thesis: "Unity ECC: Unified Memory Protection Against Bit and Chip Errors" (Advisor: Prof. Jungrae Kim) - Teaching Assistant, "Digital System Design" (Spring 2022) [SSE3016-41]	Suwon, Korea <i>Feb. 2022 – Feb. 2024</i>
<i>B.S., Semiconductor Systems Engineering</i> Teaching Assistant, "AI Basics & Uses" (Spring 2021) [GEDT020-I1]	<i>Mar. 2016 – Feb. 2022</i>

EMPLOYMENT

UT Austin Architectures for Emerging Systems (ACES) Research Group <i>Graduate Research Assistant (GRA), Dept. of Electrical and Computer Engineering</i> - Reliable Quantum Computers and Memory Systems - Developed reliable quantum computers and memory systems through research on innovative Error Correcting Codes and System Architecture.	Texas, United States <i>Aug. 2025 – Present</i>
Samsung Electronics Memory Division <i>Circuit Design Engineer</i> - HBM I/O Circuit Design - Analyzed HBM I/O Training Sequence and Eye Shmoo to handle I/O related issues such as Rx/Tx margin and Duty Cycle Adjuster (DCA) for stabilized memory systems.	Hwaseong, Korea <i>Mar. 2024 – May 2025</i>
<i>Summer Intern</i> - DRAM Rowhammer Prevention Algorithm - Implemented refresh algorithms against various DRAM row access patterns while minimizing read disturbance and power consumption.	<i>Jul. 2021 – Aug. 2021</i>
SKKU Scalable Architecture Lab <i>Research Assistant, Dept. of Semiconductor and Display Engineering</i> - Reliable Memory Systems and DRAM Microarchitecture - Developed reliable and efficient memory systems through research on innovative Error Correcting Codes and DRAM Microarchitecture.	Suwon, Korea <i>Sep. 2020 – Dec. 2023</i>

PUBLICATIONS

- T. Park, S. Gorgin, **D. Kim**, J. Shin, M. Sullivan, and J. Kim, "PoP-ECC: Robust and Flexible Error Correction against Multi-Bit Upsets in DNN Accelerators," *Design Automation Conference (DAC)*, IEEE, 2025.

- Y. Lim, **D. Kim**, and J. Kim, "SELCC: Enhancing MLC Reliability and Endurance with Single-cell Error Correction Codes," *Design, Automation & Test in Europe Conference & Exhibition (DATE), IEEE, 2024. **Best Paper Award***
- J. Lee, W. Jung, **D. Kim**, D. Kim, J. Lee, and J. Kim, "Agile-DRAM: Agile Trade-Offs in Memory Capacity, Latency, and Energy for Data Centers," *The 30th IEEE International Symposium on High-Performance Computer Architecture (HPCA), 2024.*
- **D. Kim**, J. Lee, W. Jung, M. Sullivan, and J. Kim, "Unity ECC: Unified Memory Protection Against Bit and Chip Errors," *The ACM/IEEE International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2023. **Best Student Paper Finalist, Invited to SAIF 2023, NVIDIA Research***
- W. Jung, **D. Kim**, and J. Kim, "Synergistic Integration: An Optimal Combination of On-Die and Rank-Level ECC for Enhanced Reliability," *The 20th International SoC Design Conference (ISOCC), IEEE, 2023.*
- Y. Lim, **D. Kim**, and J. Kim, "SCC: Efficient Error Correction Codes for MLC PCM," *The 20th International SoC Design Conference (ISOCC), IEEE, 2023.*
- K. Kwon, **D. Kim**, S. Park, and J. Kim, "EPA ECC: Error-Pattern-Aligned ECC for HBM2E," *International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC), IEEE, 2023.*
- **D. Kim***, Y. Lim*, S. Han, and J. Kim, "DNN Retraining Method Reducing Accuracy Degradation in Packet-Lossy Environments," *Journal of Korean Institute of Information Scientists and Engineers (KIISE), Vol. 50, No. 3, 2023.*
(* equal contributions)
- **D. Kim** and J. Kim, "YOCO: Unified and Efficient Memory Protection for High Bandwidth Memory," *The 19th International SoC Design Conference (ISOCC), IEEE, 2022.*

AWARDS & HONORS

Awards

DATE 2024 Best Paper Award [Link]	Mar. 2024
SC 2023 Best Student Paper Nomination [Link]	Nov. 2023
Fall 2023 CICE Superior Research Award (Second Place) [Link]	Sep. 2023 - Feb. 2024

Invited Presentation

Samsung AI Forum (SAIF) – Presented a poster on outstanding research in Computer Engineering.	Nov. 2023
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Scholarships

UT Austin Engineering Fellowship	Aug. 2025 – 2029
Samsung Electronics Industry-Academia Scholarship (full tuition & stipend)	Mar. 2020 – Feb. 2024
Undergraduate Research Program TA Scholarship	Apr. 2023, Aug. 2023
Director's Recommendation Scholarship	Nov. 2021
Undergraduate Research Program Student Success Scholarship	May 2021
Merit-based Semiconductor Education Scholarship	Aug. 2019
Samsung Semiconductor Scholarship (full tuition, 8 semesters)	Mar. 2016 – Feb. 2022

PATENTS

- J. Kim and **D. Kim**, "Code Generation Method, Error Correction Code Generation Apparatus, and Storage Medium Storing Instructions to Perform Code Generation Method," US Patent 12,413,247, 2025.
- J. Kim and **D. Kim**, "Method and Apparatus for Generating Code for Single Symbol Error Correction and Double Error Correction," KR Patent 10-2656075, 2024.

EDUCATIONAL RESOURCE

ECC-ExerSim (Error-Correcting Code Exercise and Simulator) [GitHub]	Sep. 2023
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- Korea Copyright Commission, No. C-2023-043210
- Study the fundamentals of Error Correcting Codes through practical experimentation and develop experimental methodologies to evaluate the reliability of memory systems based on these principles.

DRAM-exercise [GitHub]	Sep. 2023
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- Understand the DRAM Basics (organization, operation, timing parameters, refresh, rowhammer, etc)

PROFICIENCY IN SKILLS

Languages: C/C++, Python, Verilog, System Verilog

Technologies: TensorFlow, PyTorch, LaTeX

Standards: DDR, HBM

Tools: Vivado, VCS (GitHub), Verdi

LEADERSHIP EXPERIENCE

Military Service

Seoul, Korea

Honorary Discharge as a Sergeant, Auxiliary Police 112 Strike Force

Apr. 2017 – Dec. 2018

- Receive the Commendation from the Commissioner of the Seoul Metropolitan Police Agency.

Oct. 2017